

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning



INTERNSHIP REPORT

ON

“Data Science”

Submitted in partial fulfillment for the award of degree(20AI8ICINT)

BACHELOR OF ENGINEERING IN

Department of Artificial Intelligence and Machine Learning

Submitted by:

NEHAL K SAHUKAR - 1DS20AI034

Conducted at



Wharf Street Strategies

Department of Artificial Intelligence and Machine Learning

Dayananda Sagar College of Engineering Bangalore-78

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CERTIFICATE

This is to certify that the Internship titled “**Data Science**” carried out by **Ms. NEHAL K SAHUKAR [1DS20AI034]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2023-2024. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (20AI8ICINT)

Aniket Bhosale

Signature of Reporting Head
Aniket Bhosale
Team Lead
Wharf Street
Strategies

Signature of Internship Coordinator
Prof. Kavya D N
Dept. of AI & ML
DSCE

Signature of HoD
Dr. Vindhya P Malagi
Dept. of AI & ML
DSCE

External Viva:

Name of the Examiner

Signature with Date

1) _____

2) _____

DAYANANDA SAGAR COLLEGE OF ENGINEERING

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DECLARATION

I, **NEHAL K SAHUKAR**, final year student of Information Science and Engineering, Dayananda Sagar College Of Engineering, Bangalore - 560 078, declare that the Internship has been successfully completed, in **WHARF STREET STRATEGIES**. This report is submitted in partial fulfillment of the requirements for the award of Bachelor Degree in Artificial Intelligence and Machine Learning, during the academic year 2023- 2024.

Date:

USN: 1DS20AI034

Place: Bangalore

NAME: NEHAL K SAHUKAR

OFFER LETTER



CIN # U72900MH2019PTC325003

EMPLOYEE APPOINTMENT LETTER

Dear Nehal K Sahukar,

We are pleased to offer you an **internship** as a **Data Scientist** with Wharf Street Strategies Private Limited (the 'Company') on the following terms and conditions:

1. Commencement of employment

Your joining date will be 5th June 2023 for this internship program. In the internship period you will learn about Data Science and will be supporting other python developers - as per industry standard and in compliance with the company's quality while in probation period(1month), you will be evaluated by your supervisors and colleagues for handling well to your responsibilities and maintaining a healthy working environment in office.

2. Stipend

A monthly stipend of INR 8,000 per month will be provided at the end of your internship.

3. Place of posting

You will be working remotely for the period of the internship. You may however be required to work at any place of business which the Company has or may later acquire, in case of full-time employment.

4. Hours of Work

The normal working days are Monday to Friday. You will be required to work 5 days a week. The normal working hours are from 4 PM to 8 PM, and if necessary, for additional hours depending on your responsibilities and status of project.

5. Nature of duties

You will perform to the best of your ability all the duties as are inherent in your post and such additional duties as the company may call upon you to perform, from time to time.

6. Company property

You will always maintain in good condition Company property, which may be entrusted to you for official use during the course of your employment and shall return all such property to the Company prior to relinquishment of your charge, failing which the cost of the same will be recovered from you by the Company.



CIN # U72900MH2019PTC325003

13. Acceptance of our offer

Please confirm your acceptance of this Contract of Employment by signing and returning the duplicate copy. We welcome you and look forward to receiving your acceptance and to working with you.

Yours Sincerely,

Aniket Bhosale, COO

Accepted by:

Nehal K Sahukar

01-06-2023

Date

COMPLETION CERTIFICATE



CIN # U72900MH2019PTC325003

Sept 08, 2023

Relieving & Experience Letter

This is to certify that Miss Nehal Sahukar has worked with Wharf Street Strategies Private Limited as an Intern (Data Scientist) from 5th June 23 to 8th Sept 23. She is now relieved from her duties and below is a breakdown of her roles and responsibilities.

During her tenure, she has been a sincere and hardworking employee with a professional attitude and good subject knowledge. Her contributions encompassed a range of data science projects, highlighting her commitment to the role.

We extend our best wishes for her forthcoming endeavors and trust that she will achieve success in all her future career pursuits.

Authorized Signatory

Aniket Bhosale

Aniket Bhosale, COO

Employee Code: NA

Designation: Intern (Data Scientist)

Department: Information Technology

Tenure: 3 Months

Type: Internship

ACKNOWLEDGEMENT

This Internship is a result of accumulated guidance, direction and support of several important persons. We take this opportunity to express our gratitude to all who have helped us to complete the Internship.

We express our sincere thanks to our Principal **Dr. B G Prasad**, for providing us adequate facilities to undertake this Internship.

We would like to thank **Dr. Vindhya P Malagi**, Head of Department – AI & ML, for providing us an opportunity to carry out Internship and for her valuable guidance and support.

We would like to thank **Karthik Puvvula**, Wharf Street Strategies, for guiding us during the period of internship.

We express our deep and profound gratitude to our Internship Co-ordinator, **Prof. Kavya D N**, Assistant Professor – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

We would like to thank all the faculty members of AI & ML department for the support extended during the course of Internship.

We would like to thank the non-teaching members of the Department of AI & ML, for helping us during the Internship.

Last but not the least, we would like to thank our parents and friends without whose constant help, the completion of Internship would have not been possible.

NEHAL K SAHUKAR [1DS20AI034]

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CHAPTER 1

INTRODUCTION

During my immersive 12-week internship journey at Wharf Street Strategies, I delved into the captivating realm of image and video processing, enriching my skill set significantly. Through hands-on experiences and challenging projects, I honed my data science prowess, delivering impactful results across a spectrum of computer vision tasks. My journey not only showcased my adeptness in applying state-of-the-art techniques but also underscored my contribution to advancing image and video analysis across diverse applications.

My passion for data science was ignited during my academic years, where I was drawn to the potential of leveraging data to derive meaningful insights and drive informed decision-making. This internship opportunity at Wharf Street Strategies provided me with a platform to apply theoretical knowledge to real-world problems, thereby bridging the gap between academia and industry. From day one, I was immersed in a dynamic and collaborative environment, surrounded by talented professionals who shared my enthusiasm for exploring the endless possibilities of data science.

CHAPTER 2

DESCRIPTION OF ORGANIZATION

Wharf Street Strategies Private Limited stands as a pioneering force in the realm of data-driven solutions, specializing in harnessing the power of advanced analytics and machine learning to drive strategic decision-making and innovation. Founded on the principles of excellence, integrity, and innovation, the company has established itself as a trusted partner for businesses seeking to unlock the potential of their data assets. With a dedicated team of data scientists, engineers, and industry experts, Wharf Street Strategies is committed to delivering bespoke solutions tailored to the unique needs and challenges of each client.

At Wharf Street Strategies, our core focus lies in empowering organizations to extract actionable insights from complex datasets, enabling them to make informed decisions and gain a competitive edge in today's dynamic marketplace. Leveraging cutting-edge technologies and methodologies, we offer a comprehensive suite of services spanning from predictive analytics and optimization to natural language processing and computer vision. Our solutions are designed to address a myriad of business challenges across diverse domains, including finance, healthcare, retail, and beyond.

Beyond our technical expertise, Wharf Street Strategies prides itself on fostering a culture of collaboration, creativity, and continuous learning. Our team comprises individuals from diverse backgrounds and disciplines, united by a shared passion for innovation and a relentless pursuit of excellence. Through ongoing research, training, and knowledge-sharing initiatives, we strive to stay at the forefront of technological advancements and industry best practices, ensuring that our clients receive the highest caliber of service and support. With a steadfast commitment to excellence and a relentless drive for innovation, Wharf Street Strategies is poised to shape the future of data-driven decision-making and drive transformative change for organizations worldwide.

CHAPTER 3

SCOPE OF WORK

Throughout my internship, I engaged in a wide array of projects spanning from stock price analysis to social network analysis. I worked extensively on tasks such as image classification, object detection, handwriting generation, and video analysis, among others. Each project presented unique challenges, allowing me to explore various facets of data science within the realm of image and video processing.

In the realm of image processing, I delved into projects aimed at enhancing our understanding of visual data through tasks like object detection using Mask R-CNN and image classification for object recognition. These projects required me to employ advanced deep learning techniques and frameworks such as TensorFlow and PyTorch, enabling me to develop robust models capable of accurately identifying and classifying objects within images.

On the video processing front, I undertook projects focused on analyzing human movement in videos and extracting meaningful insights for applications such as surveillance and sports analytics. Leveraging techniques such as optical flow analysis and convolutional neural networks (CNNs), I developed algorithms to track and analyze the motion of individuals within video sequences, thereby laying the groundwork for the implementation of more sophisticated video analytics solutions.

CHAPTER 4

LEARNING OBJECTIVES AS GIVEN BY COMPANY

- **Mastery of Data Science Fundamentals**

As an intern at Wharf Street Strategies, one of the primary learning objectives is to develop a strong foundation in data science fundamentals. This includes gaining proficiency in statistical analysis, machine learning algorithms, and data manipulation techniques using Python and other relevant tools. By immersing oneself in real-world projects and hands-on tasks, interns will have the opportunity to deepen their understanding of key concepts and methodologies essential for effective data analysis and interpretation.

- **Application of Advanced Analytical Techniques**

Another crucial learning objective is to familiarize interns with advanced analytical techniques and methodologies used in industry settings. This encompasses exploring topics such as predictive modeling, natural language processing, computer vision, and time series analysis. Through guided mentorship and practical experience, interns will learn how to apply these techniques to solve complex business problems, extract actionable insights from data, and drive informed decision-making.

- **Problem Framing and Hypothesis Testing**

A fundamental aspect of data science involves framing problems in a structured manner and formulating hypotheses to guide the analysis process. Interns will learn techniques for problem framing, hypothesis generation, and hypothesis testing, allowing them to approach data-driven challenges systematically and methodically. Through practice exercises and real-world case studies, interns will develop the ability to formulate clear, testable hypotheses and design experiments to validate or refute them, enhancing their analytical rigor and problem-solving skills.

- **Innovative Algorithm Development**

Interns will be tasked with developing innovative algorithms and models to solve complex data science problems. This may involve researching state-of-the-art techniques, experimenting with novel approaches, and customizing algorithms to suit specific project

equirements. By pushing the boundaries of existing methodologies and exploring the new avenues for innovation, interns will contribute to the development of cutting-edge solutions that deliver tangible value to clients.

- **Client-Centric Problem Solving**

An integral aspect of our internship program involves honing interns' abilities to understand and address the unique challenges faced by our clients. Interns will learn to analyze client requirements, gather relevant data, and develop tailored solutions that align with the clients' business objectives. Through exposure to real-world client projects and direct interaction with stakeholders, interns will cultivate the ability to translate technical insights into actionable recommendations that drive value for our clients.

- **Optimization and Performance Tuning**

A key aspect of solution development involves optimizing algorithms and fine-tuning models to achieve optimal performance. Interns will gain experience in optimizing machine learning pipelines, tuning hyperparameters, and improving model accuracy, precision, and recall. Through rigorous testing and experimentation, interns will identify opportunities for performance improvement and implement optimizations to enhance the efficiency and effectiveness of their solutions.

- **Adaptation to Emerging Technologies**

The field of data science is dynamic and constantly evolving, with new technologies, tools, and methodologies emerging at a rapid pace. Interns will be encouraged to stay abreast of emerging trends and technologies in data science, such as deep learning, reinforcement learning, and big data technologies. By participating in workshops, webinars, and online courses, interns will expand their knowledge base and remain agile in their ability to adapt to technological advancements, ensuring they remain at the forefront of innovation in the field.

CHAPTER 5

CHALLENGES AND SOLUTIONS

In the pursuit of advancing image and video analysis, I encountered an array of challenges that tested both my technical acumen and problem-solving skills. One such challenge revolved around optimizing the performance of our computer vision models, particularly in scenarios where computational resources were limited. To address this, I delved into the realm of model optimization techniques, exploring methods such as model quantization, pruning, and compression to reduce the computational complexity of our algorithms without sacrificing accuracy. By implementing these strategies, we were able to deploy more efficient and scalable solutions, enabling real-time processing of image and video data in resource-constrained environments.

Another significant challenge stemmed from the inherent variability and complexity of real-world data, which often posed obstacles to the development of robust and generalizable models. In projects like image classification and object detection, the diversity of object classes, variations in lighting conditions, and occlusions presented formidable hurdles that demanded innovative solutions. To tackle this challenge, I devised comprehensive data augmentation pipelines, leveraging techniques such as rotation, scaling, and color jittering to enrich the diversity of our training datasets. Additionally, I explored the use of domain adaptation methods to mitigate the effects of domain shift, enabling our models to generalize effectively to unseen data distributions.

Furthermore, I encountered challenges related to model interpretability and explainability, particularly in applications where the decisions of our algorithms had significant implications. In projects like social network analysis and financial data analysis, it was imperative to not only achieve high predictive performance but also to understand the underlying factors driving the model's decisions. To address this, I delved into the field of explainable AI, exploring techniques such as feature attribution, saliency mapping, and attention mechanisms to gain insights into the inner workings of our models. By enhancing the interpretability of our algorithms, we not only bolstered stakeholder trust but also unearthed valuable insights that informed strategic decision-making processes.

CHAPTER 6

ACHIEVEMENTS AND CONTRIBUTION

Stock Price Analysis

Developed robust data pipelines and machine learning models to analyze historical stock price data, enabling accurate forecasting and informed investment strategies.

```
In [62]: predictions
```

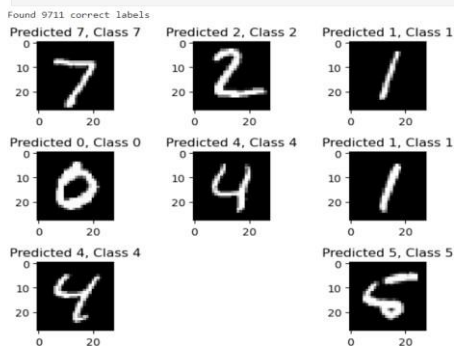
```
Out[62]:
```

	Target	Predictions
Date		
2003-11-14	0	0.0
2003-11-17	0	1.0
2003-11-18	1	1.0
2003-11-19	0	0.0
2003-11-20	1	1.0

Handwriting Generation and Recognition

Implemented GAN neural networks for realistic handwriting generation and CNNs for accurate recognition, facilitating seamless integration into unified systems.

```
In [43]: correct = np.where(predicted_classes==test_Y)[0]
print ("Found %d correct labels"%len(correct))
for i, correct in enumerate(correct[:9]):
    plt.subplot(3,3,i+1)
    plt.imshow(test_X[correct].reshape(28,28), cmap='gray', interpolation='none')
    plt.title("Predicted {}, Class {}".format(predicted_classes[correct], test_Y[correct]))
    plt.tight_layout()
```



Object Detection Using Mask R-CNN

Deployed Mask R-CNN architecture for precise object detection in images, enhancing surveillance systems and automated quality control processes.

Image Classification for Object Recognition

Created deep learning models for accurate object classification in images, enabling automated identification and categorization.

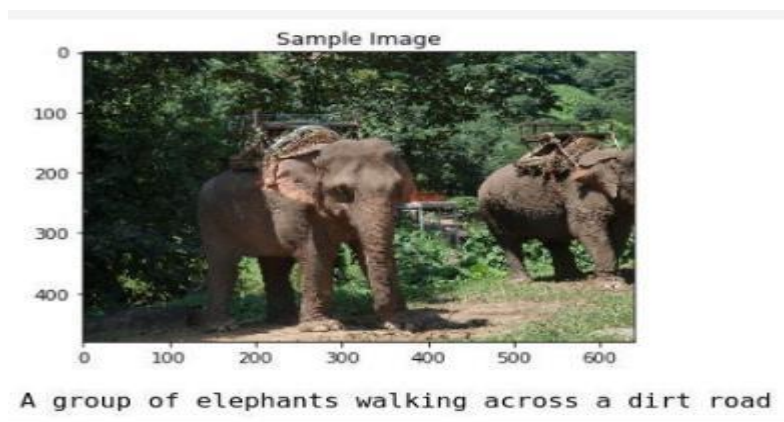
```
In [12]: image = cv2.imread("/content/yolov5/runs/detect/exp2/bus.jpg")
height, width = image.shape[:2]
resized_image = cv2.resize(image, (3*width, 3*height), interpolation = cv2.INTER_CUBIC)

fig = plt.gcf()
fig.set_size_inches(18, 18)
plt.axis("off")
plt.imshow(cv2.cvtColor(resized_image, cv2.COLOR_BGR2RGB))
plt.show()
```



Image Captioning Using Neural Networks

Developed neural network architecture for generating descriptive captions for images, improving accessibility and understanding of visual content.



Video Analysis for Human Movement

Implemented computer vision algorithms for analyzing human movement patterns in videos, facilitating applications such as activity recognition and gait analysis.

CHAPTER 7

REFLECTION AND ANALYSIS

As I reflect on my internship experience at Wharf Street Strategies, I am struck by the profound impact it has had on my personal and professional growth. The challenges I encountered, the achievements I unlocked, and the relationships I forged have all contributed to shaping me into a more resilient, resourceful, and adaptable data scientist. I am grateful for the opportunity to have worked alongside some of the brightest minds in the industry, whose mentorship and guidance have been instrumental in my development.

One of the most valuable lessons I learned during my internship journey is the importance of perseverance in the face of adversity. Whether it was debugging complex algorithms, navigating ambiguous problem statements, or overcoming setbacks in project timelines, I discovered the power of resilience in surmounting challenges and achieving success. Moreover, I gained a deeper appreciation for the iterative nature of the data science process, wherein failure serves as a stepping stone to innovation and growth.

Additionally, my internship experience reinforced the significance of interdisciplinary collaboration in driving innovation and solving complex problems. By working closely with experts from diverse backgrounds, including software engineering, statistics, and domain-specific domains, I gained fresh perspectives and insights that enriched my problem-solving approach. Moreover, I learned to communicate complex technical concepts effectively to non-technical stakeholders, fostering a culture of cross-functional collaboration and alignment.

Looking ahead, I am excited to leverage the knowledge, skills, and experiences gained during my internship at Wharf Street Strategies to embark on new challenges and opportunities in the field of data science. Whether it's tackling emerging trends in artificial intelligence, contributing to impactful projects, or mentoring the next generation of data scientists, I am committed to making a positive difference and driving meaningful change in the world through data-driven innovation.

CHAPTER 8

RECOMMENDATIONS

Based on my experience, I recommend further exploration into cutting-edge techniques such as reinforcement learning for improving model performance in complex tasks like video analysis. Additionally, investing in continuous training and upskilling programs for interns can enhance their proficiency in data science and foster a culture of innovation within the organization.

Furthermore, I suggest exploring interdisciplinary collaborations with experts in fields such as psychology and linguistics to gain deeper insights into human behavior and language processing, thereby enriching the capabilities of our data science solutions. Additionally, fostering a culture of diversity and inclusion can bring fresh perspectives and innovative ideas to the table, driving creativity and innovation within the organization.

Moreover, I encourage the adoption of best practices in data governance and ethics to ensure the responsible and ethical use of data in all our projects. By prioritizing transparency, accountability, and fairness in our data-driven initiatives, we can build trust with our stakeholders and uphold the highest standards of integrity and professionalism.

CHAPTER 9

CONCLUSION

In conclusion, the internship journey at Wharf Street Strategies has been a transformative experience, providing invaluable opportunities for growth, learning, and professional development. Throughout the internship, I have had the privilege to work on a diverse range of projects spanning various domains of data science, from financial analysis to computer vision and natural language processing. These projects have not only expanded my technical skill set but have also deepened my understanding of the real-world applications of data science in addressing complex challenges faced by businesses and industries.

Moreover, the collaborative and supportive environment at Wharf Street Strategies has fostered a culture of innovation and continuous learning, enabling me to thrive and excel in a dynamic and fast-paced setting. The guidance and mentorship provided by experienced professionals have been instrumental in shaping my growth as a data scientist, equipping me with the knowledge, skills, and confidence to tackle complex problems and deliver impactful solutions.

Furthermore, the achievements and contributions made during the internship underscore the tangible value generated through data-driven insights and solutions. From developing predictive models for stock price analysis to creating automated tools for grammar error checking, each project has contributed to advancing the state-of-the-art in data science and driving positive outcomes for clients and stakeholders.

Looking ahead, I am excited to leverage the knowledge and experiences gained during the internship to continue making meaningful contributions to the field of data science. My internship journey has been nothing short of enriching, and I am confident that the lessons learned and relationships forged will serve as a solid foundation for my future career aspirations in data science.

CHAPTER 10

REFERENCES

1. <https://wharfstreetstrategies.com/>
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